

My passion for AI sparked at the 2019 Google’s Explore ML Workshop, where I first grasped deep learning’s potential—and its limits—igniting my drive to bridge neural and symbolic paradigms. This pursuit deepened under Dr. Amit Sheth at the University of South Carolina, where my master’s thesis on Multimodal Ingredient Substitution revealed the power of Neuro-Symbolic AI (NeSy) to enhance reasoning and interpretability.

My research foundation took root at the CKM VIGIL Lab, IIT Hyderabad, where I tackled data scarcity in medical imaging. I contributed to developing the Spatial Field Fusion Network (SFFNet) for Dental X-ray segmentation—a lightweight architecture with just 0.2M parameters trained on 193 images which was published at IEEE APSCON 2023. This demonstrated the importance of specialized, resource efficient solutions in critical domains.

Building on my IIT Hyderabad foundation, I interned as an AI Engineer at EdgeNeural.AI, diving into the practicalities of deploying deep learning models. I focused on compression techniques—distillation to transfer knowledge from large models and quantization to shrink their footprint—optimizing real-world systems like object detection and segmentation models.

Experience with Natural Language Processing Drawing on NLP’s advancements and personalization demands, I researched Personalized Response Generation with Professor Manas Gaur at the University of Maryland. We adapted RLHF into Model-based feedback, using the Electra Discriminator as a reward model to replace human evaluations with a scalable, data-driven approach. This boosted the NUBIA score by 10 points, producing empathetic, context-aware AI responses, as published in ‘K-PERM: Personalized Response Generation Using Dynamic Knowledge Retrieval and Persona-Adaptive Queries’ at AAAI Symposium 2024.

Visit at Saarland At Saarland University, as a Visiting Researcher under Professor Ingmar Weber and Dr. Ferda Ofli, I experienced a transformative phase that solidified my Ph.D. aspirations. Dr. Weber’s hands-off guidance empowered me to independently tackle satellite image super-resolution, using GANs and diffusion-based models to enhance downstream tasks like change detection and disaster analysis. I learned to explore the problem space—defining goals like improving resolution for disaster monitoring—curated a dataset of satellite imagery, and designed a series of experiments to integrate generative priors, boosting detection accuracy. This autonomy, from setting objectives to evaluating outcomes, sharpened my ability to structure research and adapt experiments iteratively.

Work with Microsoft Research At Microsoft Research India, I joined the AI4Code team under Sriram Rajamani and Akash Lal, to work on Programming with Representations (PwR) project, where I built conversational programming workflows enabling LLMs to generate custom Domain Specific Languages through user dialogue. My key contribution, PwR Studio, translated these conversations into DSLs and generated Python finite state machines via Symbolic Code—a neuro-explicit approach merging neural language understanding with symbolic execution. This open-sourced work empowered domain experts to create tailored code with minimal expertise while demonstrating the importance of rigorous evaluation. I also designed rubrics, LLM-as-judge assessments, and compilation checks to ensure reliability, learning to balance neural creativity with systematic validation.

Experience with Neuro-Explicit AI My collaboration with Dr. Amit Sheth at the University of South Carolina’s AI Institute profoundly shaped my passion for neuro-symbolic AI, particularly in food computation. Another key effort was ‘Cook-Gen: Robust Generative Modeling of Cooking Actions from Recipes,’ presented at IEEE SMC 2023, where I formulated cross-modal recipe retrieval and developed cooking action recognition. The Ingredient Substitution Dataset built during my thesis has been used in the UC Irvine + Stanford Health Hackathon 2024 and submitted our work to Nature Scientific Data. These experiences taught me to leverage knowledge graphs and LLMs for robust reasoning across domains. These projects demonstrated how knowledge graphs with LLMs enable robust cross-domain reasoning,

solidifying my interest in Neuro-Explicit AI's unique synergy of symbolic interpretability and neural pattern recognition for problems requiring both accuracy and transparency.

LLM Reasoning and Knowledge Graphs Currently, as an Applied Research Scientist in the Foundational Research Team at Thomson Reuters Lab, I am advancing AI solutions for complex legal challenges. I've designed multi-agent systems that collaborate to process and analyze legal documents, enabling more sophisticated understanding than single-model approaches. My current focus is on developing Knowledge Graphs for Legal Reasoning data generation—creating structured representations of legal concepts and their relationships to enhance LLM understanding of complex legal principles. I'm also working on synthetic data generation methodologies for post-training Large Language Models, implementing Group Relative Policy Optimization for alignment and fine-tuning.